

Real-Time Construction PPE Compliance Detection Using YOLOv11: A Comparative Study with Data Augmentation Analysis and Rule-Based Compliance Classification

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Abstract—Personal Protective Equipment (PPE) compliance is a key part of job safety at construction sites. But in large-scale scenarios, the human inspection is hard because of limited human attention and the inconsistent inspection. Although automated PPE monitoring system with deep learning-based object detection has been recently developed, the existing studies primarily focus on the previous versions of the YOLO and the assessment on the safety of PPE is mostly binary. This paper aims to develop a real-time construction PPE compliance detection framework using YOLOv11 with three goals: (1) compare YOLOv11 with YOLOv8 under the same training conditions, (2) explore the effect of each augmentation technique, and (3) add a rule-based compliance classification module. In the proposed framework, the classes Person, Hardhat and Safety Vest are detected by employing a publicly available construction safety dataset. Precision, recall, mAP, F1-score, and inference speed are used to evaluate the YOLOv8n and YOLOv11n models. Besides, brightness augmentation, blur, rotation, and noise augmentation techniques are studied individually using an ablation study. The experimental results show that there is a balance between accuracy and speed, with YOLOv11n delivering higher accuracy and mAP@50-95 and YOLOv8n offering faster inference speeds. Moreover, the proposed compliance module categorizes workers to Fully compliant, Partially compliant and Non-Compliant and the accuracy of the manually verified samples is 76%. The proposed framework offers an interpretable way to monitor construction safety automatically by means of modern object detection techniques.

Keywords: Personal Protective Equipment (PPE), YOLOv11, YOLOv8, Object Detection, Construction Safety, Data Augmentation, Compliance Classification, Computer Vision

I. INTRODUCTION

The construction site is regarded as one of the toughest workplaces, as it is filled with heavy machinery, intricate processes and constantly evolving conditions. There are different safety hazards which workers may face and the use of Personal Protective Equipment (PPE) such as safety helmets and high-visibility vests are significant in preventing

work injuries. Implementing ongoing PPE compliance is still challenging however, particularly in large construction settings with multiple workers working at the same time.

The traditional safety monitoring methods primarily rely on human-based checks by supervisors. Human monitoring to detect safety violations is inefficient when it comes to large-scale construction projects because of limited observation capability, inconsistent monitoring and delayed response. Hence, an automatic monitoring system that can continuously monitor the construction environment is needed to enhance the safety management of the construction site.

With the recent advances in deep learning and computer vision, automated PPE monitoring by image and video analysis is possible. Visual information can be used to detect workers and safety equipment using object detection algorithms, which are suitable for real-time construction safety applications. Out of all the object detection frameworks, the YOLO (You Only Look Once) family has been getting a lot of attention due to its mix of detection accuracy and efficiency. The YOLO-based methods have proven to be effective for helmet and PPE detection in construction settings in several studies [1]–[4].

With the development of lightweight architectures, improved feature extraction techniques and models based on YOLO, the performance of PPE detection has been further improved in recent studies. These methods have been proven to be effective in the identification of safety helmets, vests and other protective equipment in complex construction scenarios [5], [6]. But there are still issues with implementing PPE monitoring systems in practice, including the variability of illumination, object scale, background complexity, and partial occlusion.

Several research studies have focused on advanced detection strategies and image enhancement techniques in order to increase the robustness under varying environmental conditions. In real-world monitoring systems, the ability to deal with hard visual conditions has been shown to be important for low-light detection approaches using improved YOLO architectures [7]–[9]. Likewise, data augmentation methods are often used to enhance the model’s generalization by adding brightness, blur, rotation, and image quality variations to the training data [10, 11]. But, there are many studies that look at augmentation as a combined training method instead of breaking down the characteristics of each augmentation.

While current studies have made substantial advances in the detection of PPE, there are still some drawbacks. Firstly, there is a lack of application and performance evaluation of the latest YOLO versions (YOLOv11 and later) in the field of construction PPE monitoring, as the majority of previous works are conducted on earlier versions of YOLO. To gauge the practical performance difference between YOLOv8 and YOLOv11, a direct comparison under the same experimental conditions is needed.

Secondly, the current research often combines various augmentation methods, which hinders the understanding of the specific impact of each augmentation technique on the performance of PPE detection. A systematic augmentation ablation study can offer greater insight into the robustness of the model with regard to variations in the environment.

Third, currently available PPE monitoring systems are primarily for the detection of PPE on or off. But a more detailed level of information on worker compliance is needed for practical construction safety assessment. A worker wearing only one type of PPE is in a different safety state than when wearing full protective equipment. Thus, it is essential that compliance assessment is conducted at multiple levels to make a meaningful safety evaluation.

This paper aims to overcome these drawbacks by introducing a PPE compliance detection framework using YOLOv11, which integrates object detection, augmentation analysis, and rule-based compliance classification. The main contributions of this work are summarized as follows:

- 1) .YOLOv8n and YOLOv11n are comparatively evaluated under the same training conditions to gain insights into their detection accuracy and real-time performance.
- 2) .An augmentation ablation study is performed to assess the effect of brightness, blur, rotation and noise transformations separately on PPE detection performance.
- 3) To classify the worker status according to the detected PPE items, a rule-based compliance classification module is introduced where the worker is classified as Fully Compliant, Partially Compliant, or Non-Compliant.

The rest of this paper is structured as follows. Section II explores the existing research on PPE detection using YOLO, robustness enhancement methods, augmentation techniques, and compliance evaluation approaches. In Section III, the methodology is proposed, which involves preparing the datasets, configuring the model, augmenting the data, and the compliance classification mechanism. The experimental results and discussion are presented in Section IV. Finally, Section V concludes the paper and discusses future research directions.

II. RELATED WORK

With the advancement of deep learning computer vision technology, the safety monitoring system in the construction site has been greatly improved. Existing works have primarily been related to PPE detection, enhancing detection robustness in challenging scenarios and creating lightweight PPE detection models for real-time applications. This section summarizes the current literature on PPE detection using YOLO, methods for enhancing their robustness, data augmentation methods, and compliance evaluation methods.

A. PPE Detection using YOLO.

YOLO is a series of object detection models, which has gained popularity in the field of construction safety monitoring for its ability to meet the requirements of high accuracy and low computational cost. Initial research primarily centered on automated safety helmet detection, which showed that advanced YOLO models could detect construction images and identify protective equipment while keeping the processing speed high.

Hu and Ren have introduced YOLO-LHD, a lightweight framework for helmet detection in industrial environment. The results of the study showed that lightweight models for YOLO could effectively detect helmets and also lower the burden of the calculation, which can be used in practical safety monitoring applications [1].

Some studies have also delved into the PPE detection using YOLO in construction scenarios. Helmet and PPE detection using YOLOv8 has been demonstrated to achieve better results in complex construction scenarios, with a focus on feature extraction and network optimization [2,3]. Other research has proposed adapted YOLO structures to better detect small objects, particularly for safety equipment which takes up small areas in images [4].

In recent studies, the use of the newer versions of YOLO for PPE monitoring has also been explored. Despite the enhancement of feature extraction and the efficiency of the model, the construction safety monitoring of YOLOv11 is still relatively limited compared with the previous YOLO versions. Most of the previous works review one model and not compare two or more generations of the YOLO models

under the same experimental condition.

B. Robust Object Detection for Construction Safety Monitoring.

Object detection performance may be degraded in several challenges that are introduced in real life construction environments. The different illuminations, perspectives of the cameras, working positions of the workers, scales of the objects and difference of backgrounds bring up difficulties to the automated monitoring system.

The problems have been tackled by several studies, which have worked to improve feature extraction methods and create better detection architectures. YOLO based models that are lightweight have been explored to have faster inference time while still having a good detection performance, especially for systems that are supposed to have real-time monitoring [5] [6].

One of the important challenges in construction surveillance is low-light conditions, since the light scattered in the images taken by surveillance cameras may not be enough. Recently, several YOLO-based methods have been studied to further develop the YOLO framework for low-light object detection, such as using feature enhancement or attention mechanisms. These studies illustrate how environmental variations play a key role in object detection performance and the need to improve our model's robustness [7]–[9].

But a lot of previous works are considering the general problem of object detection in image instead of construction PPE problem. So additional research needs to be done to better understand the effect of various environmental changes on PPE detection.

C. Data Augmentation on PPE Detection.

In deep learning object detection, data augmentation is often employed to expand the ability of the model to detect objects in diverse conditions. Augmentation techniques are ways to create new training samples, which can help the model learn variations that might arise in the real world.

Brightness adjustment, blur, rotation and noise injection are common augmentation techniques used. These changes mimic real-world scenarios like lighting conditions, camera movements, viewpoint variations and poor image quality.

There are several PPE detection studies which have used augmentation in the training of the models and detected that detection performance can be improved with augmentation [10, 11]. But most of the current methods use several augmentation methods at the same time and test the performance of the resulting model. This makes it difficult to determine the specific effects of each augmentation procedure.

This analysis is significant in the context of construction safety as various transformations can have varying impacts on the ability to detect PPE. Brightness variation will, for instance, make the PPE more robust to illumination changes; too much noise will cause the loss of the important visual information in the case of small PPE objects. Hence, an individual augmentation ablation study is needed to determine the most effective augmentations for construction PPE detection.

D. Compliance-Level Safety Assessment

Most of the current monitoring systems for PPE are aimed at identifying the presence or absence of certain safety equipment. While this information is valuable, binary is only partial data on the safety status of individual workers.

In the real world of construction, worker compliance is more complicated. A worker in a helmet without a safety vest is a different safety situation than a worker who is wearing a full safety vest. Hence, it is possible to create a multi-level compliance assessment system that can give more meaningful information for safety management.

Automated PPE monitoring and safety violation detection systems have been investigated in the past [12] [13]. But, many of the current solutions only generate detection results or violation alerts and do not give detailed compliance categories for workers.

Compliance classification by detected PPE items can help to increase the interpretability of automated safety monitoring systems. In this research, we aim to overcome this limitation by developing a rule based compliance module for workers which classifies them in Fully Compliant, Partially Compliant and Non-Compliant classes.

E. The discussion below addresses some of the current limitations.

From the literature reviewed there are some gaps for research. Past PPE detection research has primarily concentrated on the previous YOLO models, and there has been a lack of extensive analysis of the performance of newer YOLO models for construction safety monitoring. Moreover, the effect of each of the augmentation techniques is not well investigated as most of the studies employ combined augmentation techniques.

Moreover, most of the current monitoring systems are geared towards the detection of PPE and do not offer any worker-level compliance assessment. These constraints inspire the suggested framework that integrates YOLOv11 detection, controlled augmentation analysis, and rule-based compliance classification.

III. METHODOLOGY

To achieve automated construction PPE monitoring, the proposed framework combines YOLO-based object detection with

augmentation analysis, and a rule-based compliance assessment module. The entire process is divided into four phases: the preparation of the dataset, the detection of the objects, the assessment of the augmentation and the classification of the compliance of the workers.

A. Proposed Framework

The overall architecture of the proposed system is shown in Fig. 1.

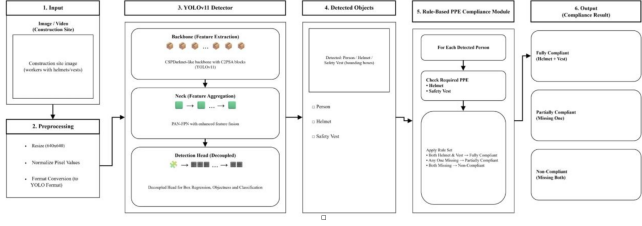


Fig. 1. The overall architecture of the proposed PPE compliance detection framework using YOLOv11.

The YOLO detector is first used to process the input image or video frame and detect three target classes: Person, Hardhat, and Safety Vest. The detected objects and their bounding box information is then fed to the compliance classification module. The system categorizes compliance by one of three levels based on the PPE detected for each worker: Fully Compliant, Partially Compliant, or Non-Compliant.

The proposed framework offers an interpretable worker-level safety assessment without an extra classification network, which is not possible with the traditional PPE detection methods that only detect the presence or absence of safety equipment.

B. Dataset Description

The experiments are performed on a publicly available Construction Site Safety dataset from Roboflow Universe. The dataset includes labelled construction images with YOLO format.

The original data set contains several classes of objects but this study concentrates on three classes of objects directly related to PPE compliance:

- 1) Person
- 2) Hardhat
- 3) Safety Vest

The data set contains 2,801 images, 2,603 of which are used for training and 114 for validation. Before model training, all the images are resized to 640×640 pixels.

The chosen data set contains the bounding boxes for each object, which can be used to train YOLO-based object detection models.

TABLE I
DATASET SUMMARY

Property	Value
Total images	2,801
Training images	2,603
Validation images	114
Input resolution	640×640
Target classes	Person, Hardhat, Safety Vest

C. Model Configuration and Training

In this study, the two lightweight object detection models, YOLOv8n and YOLOv11n, are evaluated. The nano variants are chosen for their ability to offer a balance between precision and efficiency for real-time safety monitoring.

Both models are trained with the same experimental settings to make a fair comparison. The models are pre-trained with COCO weights and fine-tuned on the construction PPE dataset.

TABLE II
TRAINING CONFIGURATION

Property	Value
Parameter	Setting
Models	YOLOv8n, YOLOv11n
Image size	640×640
Epochs	50
Batch size	16
Optimizer	AdamW
GPU	NVIDIA Tesla T4

Precision, recall, F1-score, mAP@50, mAP@50-95, inference time and FPS are used to assess the model performance.

D. Augmentation Ablation Strategy

To study the impact of various image transformations on PPE detection, data augmentation is performed. This study aims to assess the effect of each augmentation individually by conducting an ablation study.

Four augmentation techniques are investigated:

Brightness Adjustment: Represents various light situations that are typical for outdoor construction sites.

Blur Augmentation: Models camera motion or image degradation due to focus.

Rotation Augmentation: Models different camera positions and worker orientations.

Noise Augmentation: Are poor quality pictures from substandard surveillance cameras.

Each experiment is conducted with a different training set created with one of the augmentation techniques and the validation set remains the same. This design enables the individual effect of each augmentation method to be measured.

E. Rule-Based Compliance Classification

Once Person, Hardhat, and Safety Vest objects have been detected, a rule-based module checks the compliance level of each worker. Bounding box overlap is used to determine the

association between a worker and a PPE item.

If the overlap ratio is above a certain threshold, the detected PPE item is assigned to a worker.

The rules of classification are as follows:

1) *Fully Compliant*: Person found wearing both Hardhat and Safety Vest.

2) *Partially Compliant*: Person found with one of the PPE items.

3) *Non-Compliant*: Person found without Hardhat and Safety Vest.

This rule-based method offers a straightforward and comprehensible compliance evaluation process without adding to the complexity of the model.

F. Evaluation Procedure

Standard object detection metrics are used to evaluate the trained models on the validation dataset. Precision and recall are used to measure correctness of detection and capability of object retrieval, and mAP values are used to measure the overall detection performance.

The models' inference speed is also tested using FPS and inference time to assess their suitability for real-time monitoring applications.

Randomly selected validation images are manually verified and compared with the compliance categories predicted by the model to compute classification accuracy for compliance evaluation.

IV. RESULTS AND DISCUSSION

The experimental evaluation of the proposed framework for PPE compliance detection is presented in this section. The performance of YOLOv8n and YOLOv11n is compared using identical training conditions. In addition, an augmentation ablation study is performed to investigate the effect of various image transformations. Finally, the performance of the proposed rule based compliance classification module is tested.

A. Detection Performance Comparison

Precision, recall, F1-score, mAP@50, mAP@50-95, inference time, and FPS are used to evaluate the performance of the YOLOv8n and YOLOv11n models. The same dataset and configuration are used to train both models, allowing for a fair comparison.

The results show that YOLOv11n outperforms YOLOv8n in terms of precision, indicating a better capability to minimize false positive detections. YOLOv11n also offers a higher mAP@50-95 score, which reflects improved localization accuracy under more stringent criteria. YOLOv8n, however, offers a better recall and faster inference speed, making it more appropriate for applications that require faster processing.

TABLE III
OVERALL PERFORMANCE COMPARISON BETWEEN YOLOv8n AND YOLOv11n

Metric	YOLOv8n	YOLOv11n
Precision	0.914	0.951
Recall	0.761	0.742
F1-score	0.831	0.834
mAP@50	0.838	0.825
mAP@50-95	0.495	0.505
Inference Time (ms)	4.5	6.8
FPS	222	147

The results show a compromise between detection accuracy and computation efficiency. Thus, the model to be used is dependent on the requirements of the target deployment environment.

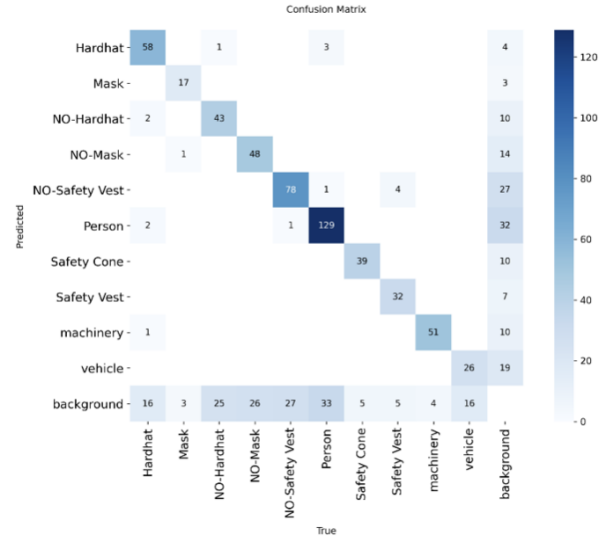


Fig. 2. YOLOv11n confusion matrix for validation set.

The confusion matrix shows the distribution of predictions made by YOLOv11n for each class. The model achieves high classification accuracy for the target classes, with most of the misclassifications being attributed to background clutter and failure to detect in cluttered scenes.

B. Per-Class Detection Analysis

Class-wise performance is also evaluated for Person, Hardhat and Safety Vest classes to further analyze the detection behavior.

TABLE IV
PER-CLASS DETECTION PERFORMANCE

Model	Class	Precision	Recall	mAP@50	mAP@50-95
YOLOv8n	Hardhat	0.954	0.747	0.826	0.506
YOLOv8n	Person	0.875	0.756	0.815	0.466
YOLOv8n	Safety Vest	0.914	0.779	0.872	0.513
YOLOv11n	Hardhat	0.967	0.731	0.816	0.494
YOLOv11n	Person	0.917	0.693	0.799	0.496
YOLOv11n	Safety Vest	0.970	0.801	0.861	0.525

Safety Vest provides the best detection performance for both models of the three target classes. YOLOv11n delivers improved accuracy for Hardhat and Safety Vest detection, and YOLOv8n offers comparable recall.

The Person class still has a low recall rating, which suggests that scenes with multiple people and overlapping workers are still difficult. These challenges can impact the compliance classification, since when persons are not detected correctly, PPE may not be associated correctly.

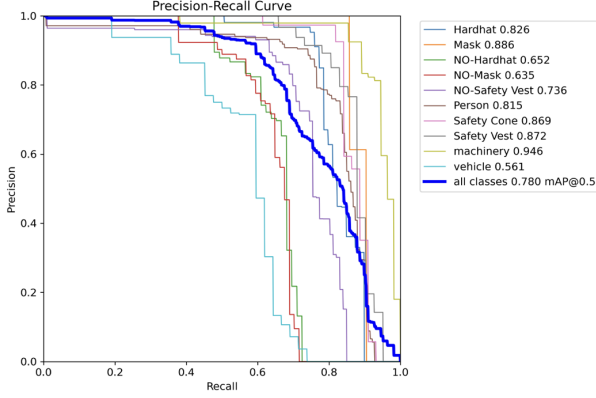


Fig. 3. Precision-recall curves for Person, Hardhat, and Safety Vest classes of YOLOv11n.

The precision-recall curves further demonstrate the detection capability of YOLOv11n for different classes. The model has good performance for Safety Vest detection, and Person detection is relatively difficult because of differences in the appearance of workers and the complexity of the scenes.

C. Augmentation Ablation Study

YOLOv11n is used to evaluate the effect of individual augmentation techniques. The performance of each augmentation method is tested individually to see its effect on detection performance.

TABLE V
AUGMENTATION ABLATION RESULTS USING YOLOV11N

Configuration	Precision	Recall	mAP@50	mAP@50-95
Baseline	0.893	0.691	0.798	0.439
Brightness	0.916	0.669	0.807	0.422
Blur	0.896	0.690	0.781	0.425
Rotation	0.805	0.596	0.688	0.237
Noise	0.671	0.636	0.652	0.272

The augmentation results indicate that the performance of the model varies with different transformations. In terms of illumination variation, mAP@50 is slightly improved with brightness adjustment compared to the baseline. The performance level is similar with blur augmentation, which has little effect.

Rotation and noise augmentation, on the other hand, decrease detection accuracy. Too much rotation can impact localization of objects and noise can strip away significant visual information from small PPE objects.

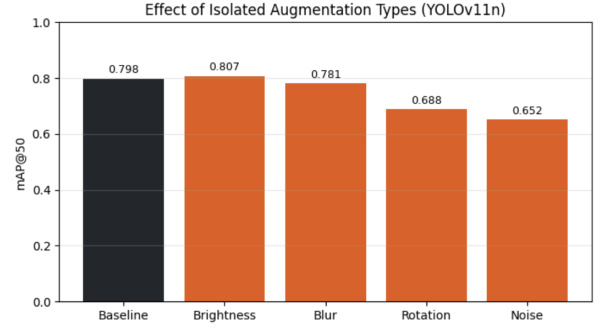


Fig. 4. Comparing mAP@50 for various augmentation setups.

The graphical comparison shows that the brightness augmentation transformation yields the best result among the transformations tested and the noise transformation yields the worst result.

D. Compliance Classification Evaluation

The proposed rule-based compliance module is tested with the manually verified samples from the validation data set.

TABLE VI
COMPLIANCE CLASSIFICATION PERFORMANCE

Metric	Value
Images evaluated	18
Worker instances	25
Correct classifications	19
Accuracy	76%

The compliance classification module is 76% accurate, showing that it is feasible to translate the object detection results to worker-level safety assessment. The rule-based approach offers an interpretable decision process without the need for further model training.

The most common sources of error are PPE not detected, PPE overlap, and PPE detected but not worn. These limitations indicate the need for more advanced worker-PPE association methods in future work.

The visualization shows how the proposed framework can recognize workers and assign different compliance levels according to the PPE items detected.

E. Discussion

The experimental results show that the performance of PPE monitoring is a trade-off between detection accuracy, processing speed and environmental robustness. YOLOv11n offers enhanced precision and localization performance, while YOLOv8n delivers faster inference speeds.

The augmentation analysis indicates that more training variations does not necessarily lead to better performance. The effectiveness of augmentation is dependent on the target environment characteristics and the type of transformation.



Fig. 5. Sample results of the proposed PPE compliance classification system.

Also, the compliance classification module is a further extension of the traditional PPE detection, which gives a more meaningful safety assessment. While the current rule-based method has some drawbacks, it shows the potential of combining the output of object detection with an interpretable decision-making method.

V. LIMITATIONS AND FUTURE WORK

While the proposed framework shows how automated PPE compliance monitoring is possible, there are still a number of limitations. First, the experiments are performed on a publicly available construction safety dataset, instead of real images collected from construction sites. Therefore, variations in actual deployment environments, such as different camera positions, weather conditions, and site layouts, may affect model performance.

Second, the existing system is based on three main classes: Person, Hardhat and Safety Vest. Other important safety equipment, including gloves, safety boots and protective goggles, are not included in this study. Expansion of the system to incorporate more categories of PPE can be used to give a more complete safety evaluation.

Third, the compliance classification module is based on a rule-based association between detected workers and PPE items. While this method allows for an interpretable decision process, it can generate errors in complex scenes with multiple workers and PPE objects.

Future research will involve testing the framework with larger and more varied construction datasets, such as actual surveillance videos from various environments. Other PPE categories will be added to enhance safety coverage. Additionally, state-of-the-art worker-PPE association methods and severity-

based compliance scoring approaches will be investigated to offer more accurate and detailed safety assessment.

VI. CONCLUSION

This paper introduced a real-time construction PPE compliance detection framework using YOLOv11, comparative analysis, augmentation evaluation, and rule-based compliance classification. The proposed system classifies Person, Hardhat, and Safety Vest classes, and can be used to augment existing PPE detection systems with worker-level compliance assessment.

YOLOv8n and YOLOv11n were compared under the same training conditions. The results showed a trade-off between accuracy and speed, with YOLOv11n outperforming in terms of accuracy and mAP@50-95, while YOLOv8n offered faster inference performance. The augmentation ablation study also revealed that the augmentation techniques have varying impacts on detection performance. Brightness variation increased the robustness of the detection, while rotation and noise decreased the detection performance when applied alone.

The proposed rule-based compliance module was tested on manually verified samples with 76% accuracy, showing the potential for combining object detection with an interpretable safety assessment. While more enhancements are needed for practical use, the proposed framework offers a solid basis for automated construction safety monitoring.

Future studies will focus on larger scale evaluation, other categories of PPE, better worker-PPE association methods, and implementation on real-time surveillance systems.

REFERENCES

- [1] L. Hu and J. Ren, "YOLO-LHD: An improved lightweight method to detect helmet wearing in industrial settings," *Frontiers in Built Environment*, vol. 9, Art. no. 1288445, 2023, doi: 10.3389/fbuil.2023.1288445.
- [2] Z. Wu, X. Lei, and M. Kumar, "Advancing construction safety: YOLOv8-CGS helmet detection model," *PLOS ONE*, vol. 20, no. 5, Art. no. e0321713, 2025, doi: 10.1371/journal.pone.0321713.
- [3] J. Han, Y. Liu, Z. Li, Y. Liu, and B. Zhan, "Safety helmet detection based on YOLOv5 driven by super-resolution reconstruction," *Sensors*, vol. 23, no. 4, Art. no. 1822, 2023, doi: 10.3390/s23041822.
- [4] A. Barlybayev, N. Amangeldy, B. Kurmetbek, I. Krak, B. Razakhova, N. Tursynova and R. Turebayeva, "Personal protective equipment detection using YOLOv8 architecture on object detection benchmark datasets: A comparative study," *Cogent Engineering*, vol.11, no.1, art.2333209, 2024, doi: 10.1080/23311916.2024.2333209.
- [5] G. Yang, X. Hong, Y. Sheng, and L. Sun, "YOLO-Helmet: A novel algorithm for detecting dense small safety helmets in construction scenes," *IEEE Access*, vol. 12, pp. 107170–107180, 2024, doi: 10.1109/ACCESS.2024.3435700.
- [6] X. Jiao, C. Li, X. Zhang, J. Fan, Z. Cai, Z. Zhou, and Y. Wang, "Detection method for safety helmet wearing on construction sites based on UAV images and YOLOv8," *Buildings*, vol. 15, no. 3, Art. no. 354, 2025, doi: 10.3390/buildings15030354.
- [7] D. Kim and S. Xiong, "Enhancing worker safety: Real-time automated detection of personal protective equipment to prevent falls from heights at construction sites using improved YOLOv8 and edge devices," *Journal of Construction Engineering and Management*, vol.151, no.1, 2025, doi: 10.1061/JCEMD4.COENG-14985.

- [8] ‘D. Zhao, F. Shao, S. Zhang, L. Yang, H. Zhang, S. Liu, and Q. Liu, “Advanced object detection in low-light conditions: Enhancements to YOLOv7 framework,” *Remote Sensing*, vol. 16, no. 23, Art. no. 4493, 2024, doi: 10.3390/rs16234493.
- [9] S.U.A. Shovo , M.G.R.Abir , M.M.Kabir , M.F. Mridha , “Advancing low-light object detection with you only look once models: An empirical study and performance evaluation,” in *Cognitive Computation and Systems*, vol. 6, no. 4, pages 9–20, 2014. 119–134, 2024, doi: 10.1049/ccs2.12114.
- [10] ‘ X. Liu, C. Liu, X. Zhou, and G. Fan, “Enhancing low-light object detection with En-YOLO: Leveraging dual attention and implicit feature learning,” *Multimedia Systems*, vol. 31, no. 3, Art. no. 249, 2025, doi: 10.1007/s00530-025-01820-7.
- [11] ‘W. Cai, Y. Chen, X. Qiu, M. Niu, and J. Li, “LLD-YOLO: A low-light object detection algorithm based on dynamic weighted fusion of shallow and deep features,” *IEEE Access*, vol. 13, pp. 69967–69979, 2025, doi: 10.1109/ACCESS.2025.3558574.
- [12] I. Makris, A. Lytos, K. Kyranou, V. Argyriou, T. Lagkas, K.-F. Kollias, G. F. Fragoulis, and P. Sarigiannidis, “Detecting personal protective equipment (PPE) utilising YOLOv8 in a federated learning environment,” *AIP Conference Proceedings*, 2024, Art. no. 050006, doi: 10.1063/5.0236071.
- [13] ‘B. Tong, G. Li, X. Bu, Y. Wang, and X. Yu, “A deep learning-based algorithm for the detection of personal protective equipment,” *PLOS ONE*, vol. 20, no. 5, Art. no. e0322115, 2025, doi: 10.1371/journal.pone.0322115.